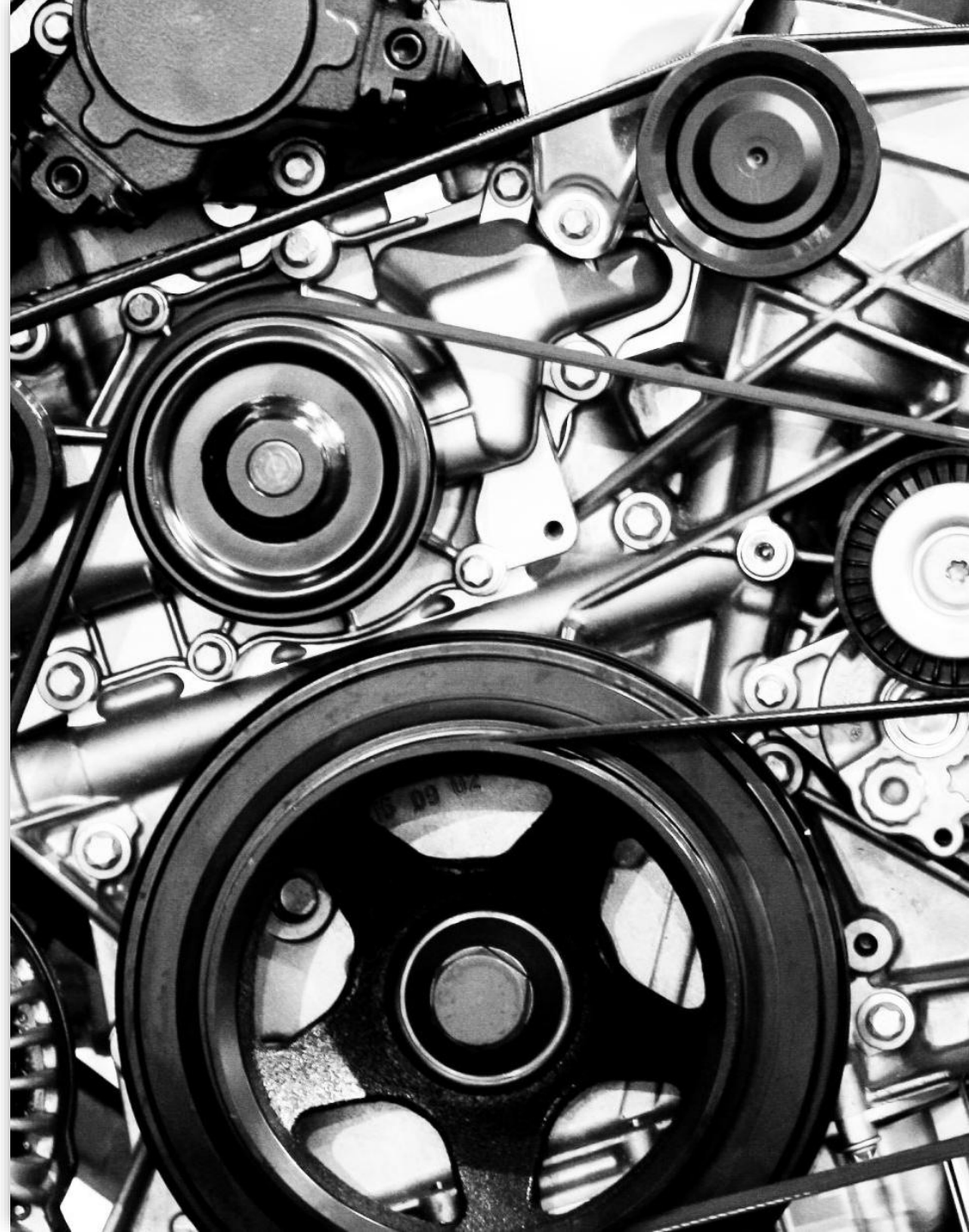


What is FastAPI?

- **The Modern Standard:** FastAPI is a high-performance web framework for building APIs with Python 3.8+ based on standard Python type hints.
- **Speed:** It is one of the fastest Python frameworks available, rivaling **Go** and **NodeJS**, thanks to its underlying **Starlette** and **Pydantic** engines.
- **Developer Experience:** * **Automatic Docs:** It automatically generates interactive API documentation (Swagger UI) as you code.
- **Type Safety:** Uses Python type hints to catch errors early and provide great editor support (autocompletion).



How It Works (The Mechanics)

- **Client-Server Bridge:** FastAPI acts as the "Server," waiting for "Clients" (like mobile apps or web browsers) to send requests. It then processes the logic and returns data, usually in **JSON** format.
- **The ASGI Advantage:** Unlike older frameworks, FastAPI is built on **ASGI** (Asynchronous Server Gateway Interface), allowing it to handle many simultaneous connections without "blocking" or waiting.
- **Deployment:** It is typically paired with **Uvicorn**, a lightning-fast server engine that hosts the FastAPI application on a local or cloud network.

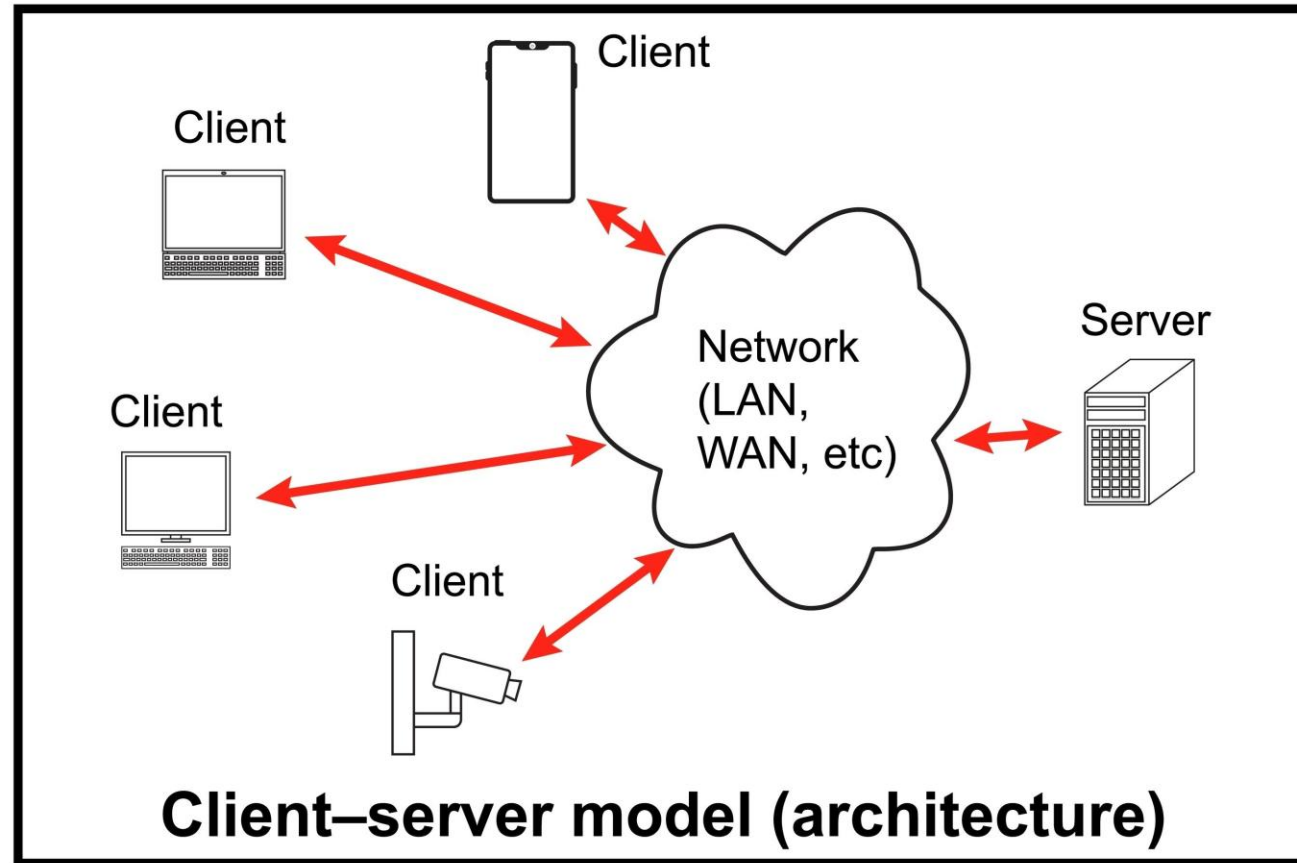


Welcome & Module Induction

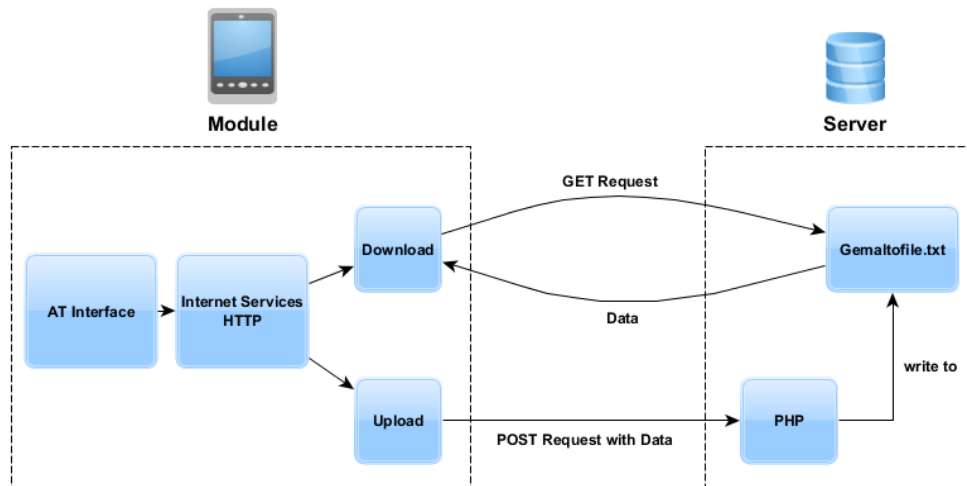
- **What we learned previously:** In Module 10, we mastered being "Clients." We used the requests library to fetch data from across the web and learned how to handle JSON payloads and authentication headers.
- **What we are learning today:** We are switching sides. We are becoming the "Server." We will learn the Client-Server architecture and build our first web endpoints using **FastAPI**.
- **Why this will be helpful:** In modern AI engineering, your model (like a 3D segmentation script) needs to be accessible to other apps. FastAPI allows you to wrap your Python logic into a "Service" that anyone with a URL can interact with. It is fast, professional, and industry-standard.

What is an API? (Client vs. Server Architecture)

- **The Client:** The application that asks for data (like a mobile app or your scripts from Module 10).
- **The Server:** The program that listens for requests, processes logic, and sends back a response (the "Service").
- **The API (Application Programming Interface):** The bridge that allows them to communicate.

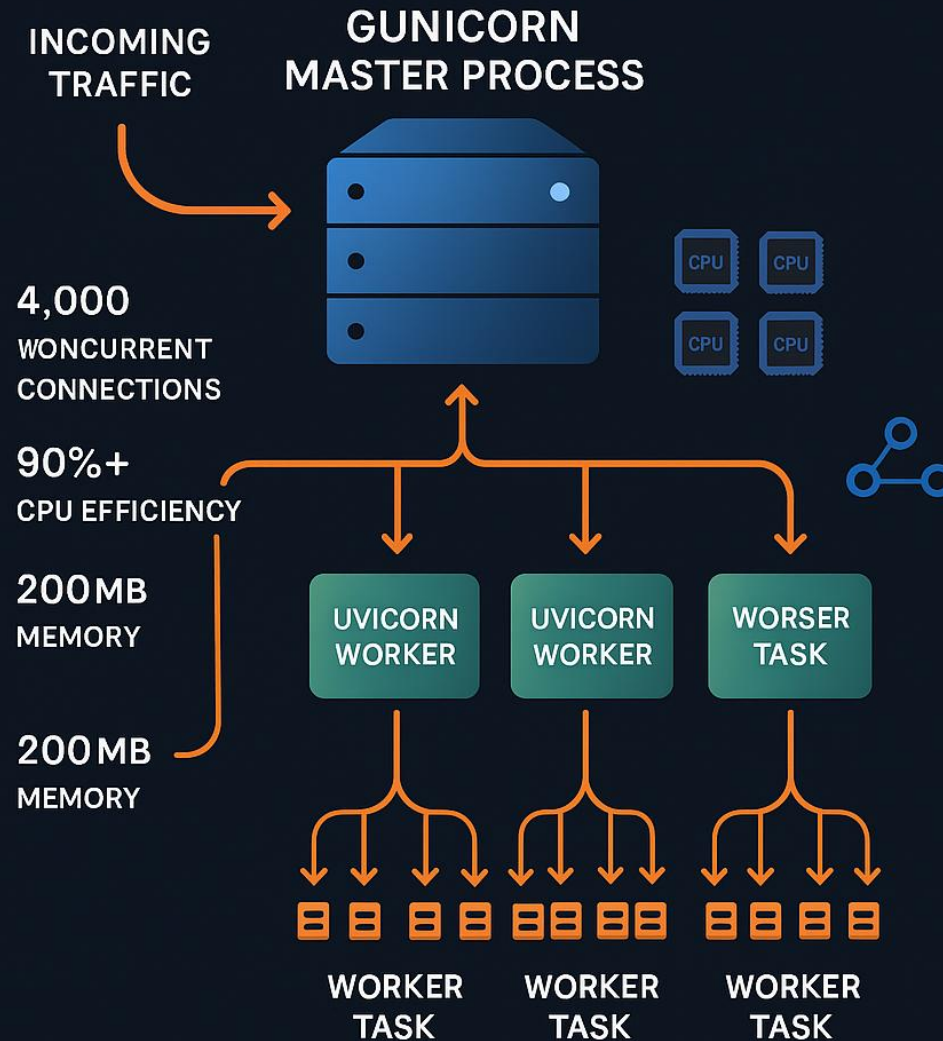


Building Your First Endpoints (GET & POST)



- **The GET Endpoint:** Used to retrieve data. For example, checking if the AI server is online.
- **The POST Endpoint:** Used to send data to the server. For example, uploading an image file for processing.
- **Automatic Docs:** FastAPI automatically generates a professional "Swagger" UI at /docs so users know exactly how to use your tool.

Uvicorn & Local Hosting



- **The Engine:** While FastAPI is the code, **Uvicorn** is the lightning-fast engine (ASGI server) that runs it.
- **Localhost:** Your computer will act as the server, listening on port 8000. You can access it at <http://127.0.0.1:8000>.